Section 6.1 Homework (Homework)

INSTRUCTOR
Rocio Owens
American River College, CA

Current Score

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
POINTS	1/1 ✓	1/1 ✓	1/1 ✓	5/5 ✓	2/2 ✓	2/2 ✓	2/2 ✓	2/2 ✓	2/2 ✓	2/2 ✓	2/2 ✓	2/2 ✓	2/2 ✓	1/1 ✓	6/6 ✓

TOTAL SCORE

33/33	100.0%

Due Date

Past Due

TUE, JAN 26, 2021
11:59 PM PST

Request Extension

Description

Assignment Submission & Scoring

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your last submission is used for your score.

The due date for this assignment has passed.

Your work can be viewed below, but no changes can be made.

Important! Before you view the answer key, decide whether or not you plan to request an extension. Your Instructor may not grant you an extension if you have viewed the answer key.

Request Extension

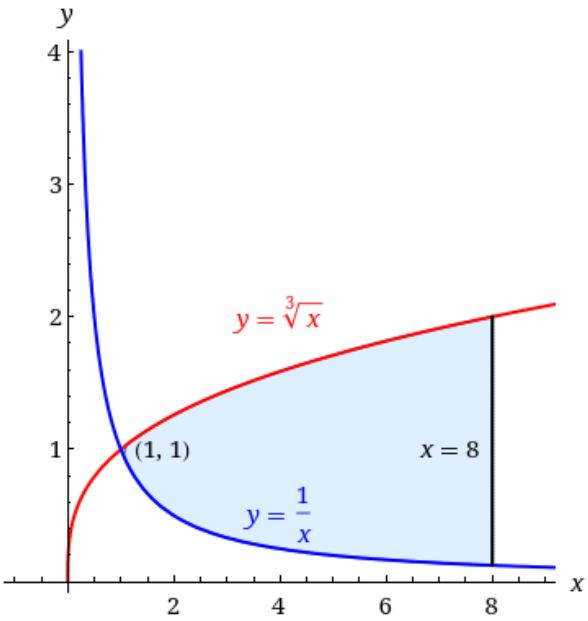
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1. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 6.1.001.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Find the area of the shaded region.

$$454−ln(8)+ln(1)



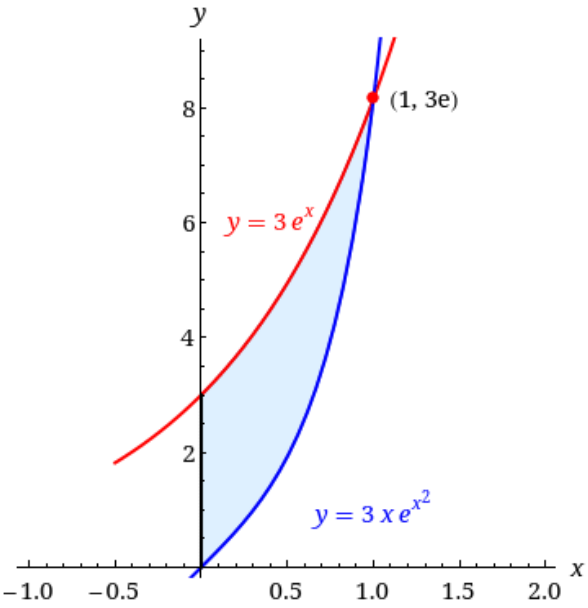
Need Help? Read It Watch It

2. [1/1 Points] DETAILS PREVIOUS ANSWERS SCALCET8 6.1.002.

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Find the area of the shaded region.

$$(3e−3)2



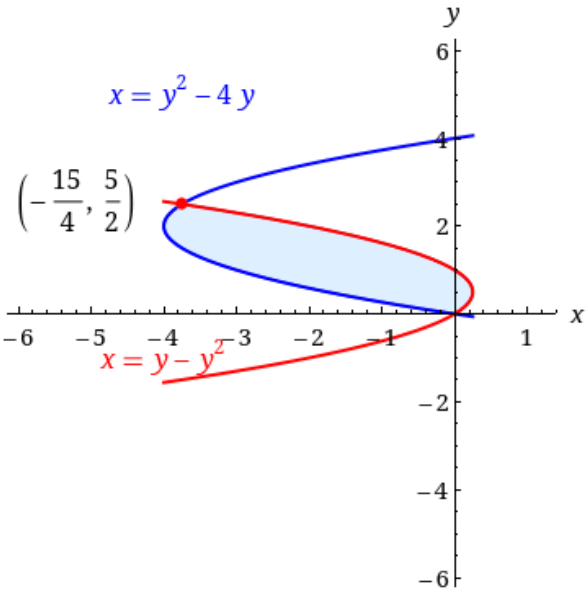
Need Help? Read It

3. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 6.1.004.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Find the area of the shaded region.

$$12524



Need Help? Read It

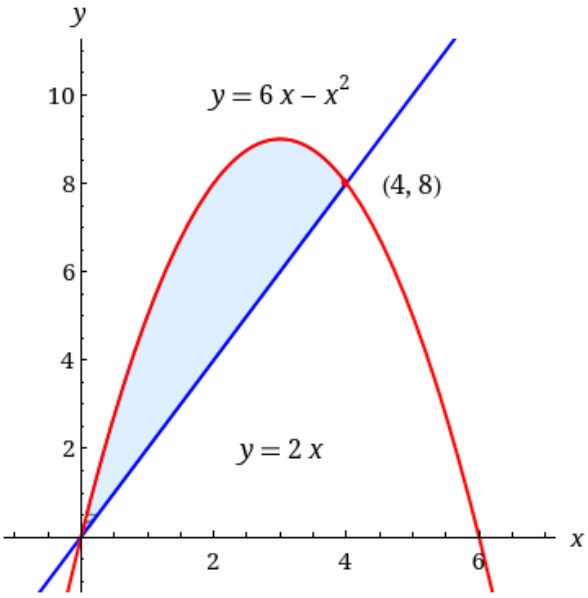
4. **[5/5 Points]** DETAILS PREVIOUS ANSWERS SCALCET8 6.1.501.XP.MI.SA.

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

This question has several parts that must be completed sequentially. If you skip a part of the question, you will not receive any points for the skipped part, and you will not be able to come back to the skipped part.

Tutorial Exercise

Find the area of the shaded region.



Step 1

To find the area between these curves, we find the integral of the "top function" minus the "bottom function" over the region, which would be the function y =

$$6x−x2

✔ minus the function y =

$$2x

✔ .

Step 2

We must integrate $y = (6x - x^2) - (2x) = 4x - x^2$.

The limits on the integral correspond to the smallest and largest possible x-values where the two graphs intersect, and so will be

$\int_0^{4}$ ✔ $(4x - x^2)\,dx$.

Step 3

$$\text{Area} = \int_0^4 (4x - x^2)\,dx = \Big[$$

$$2x2−x33

We now have ✔ $\Big]_0^4$.

Step 4

Thus, the exact area is $S =$

$$323

.

You have now completed the Master It.

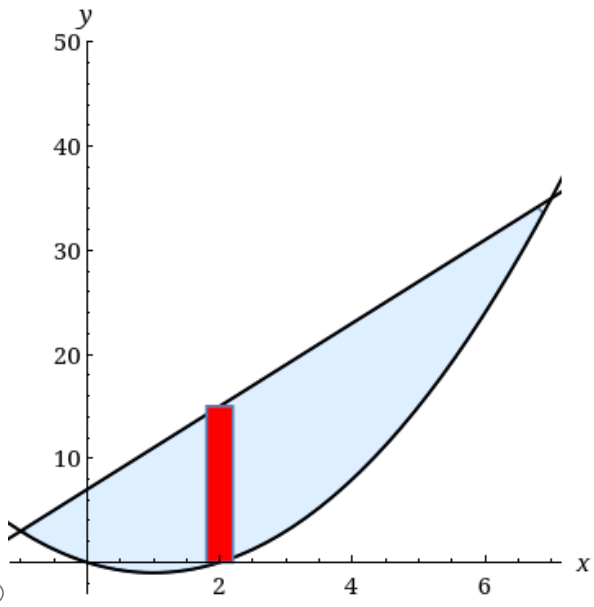
Need Help? Read It

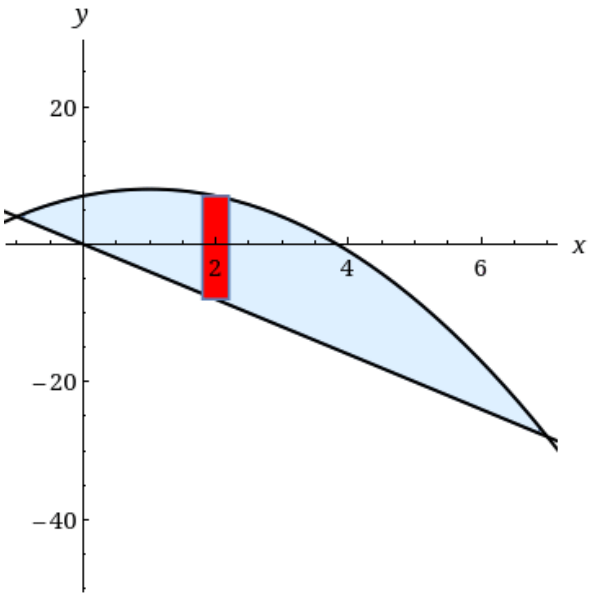
5. **[2/2 Points]** DETAILS PREVIOUS ANSWERS SCALCET8 6.1.502.XP.

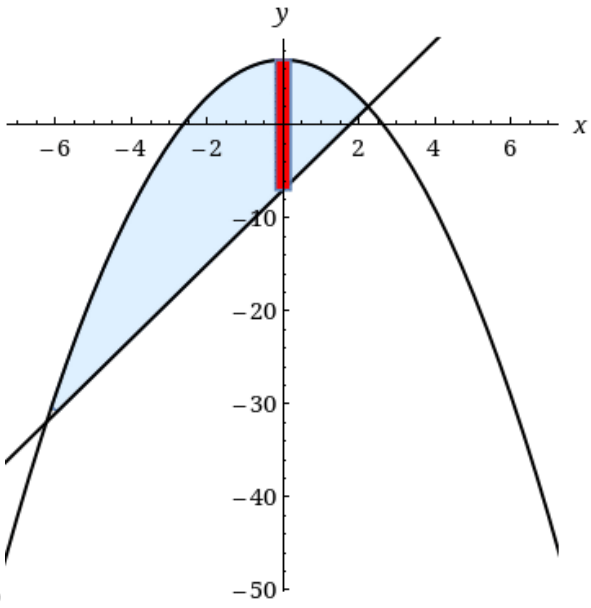
MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

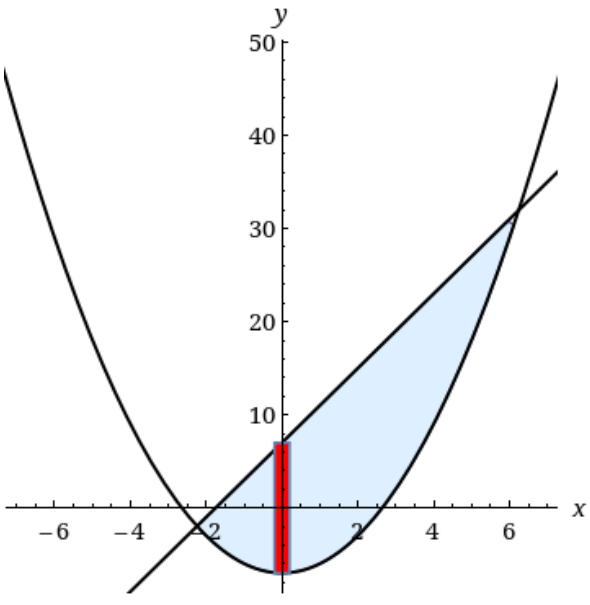
Sketch the region enclosed by the given curves. Decide whether to integrate with respect to x or y. Draw a typical approximating rectangle.

$y = x^2 - 2x, \quad y = 4x + 7$









Find the area of the region.

$$2563

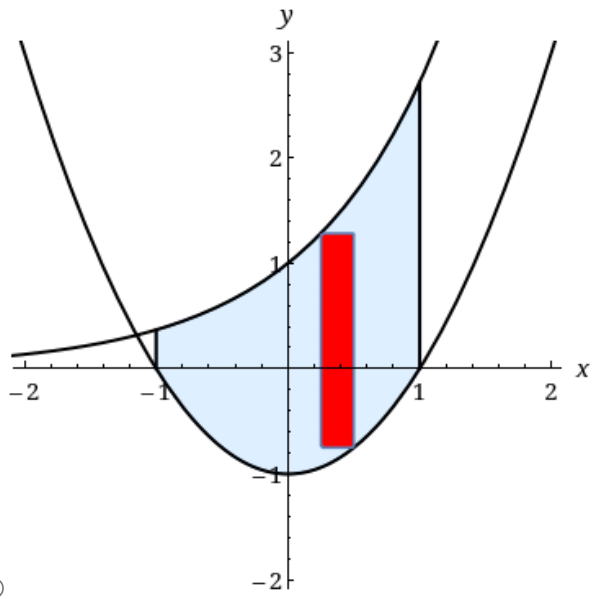
Need Help? Read It

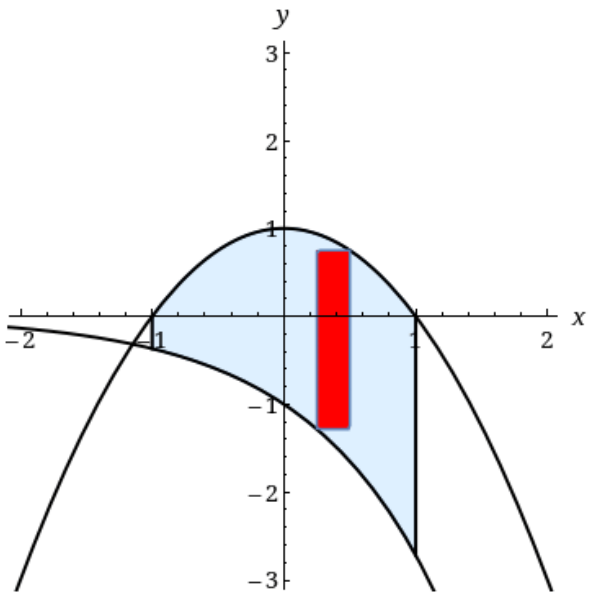
6. **[2/2 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 6.1.005.**

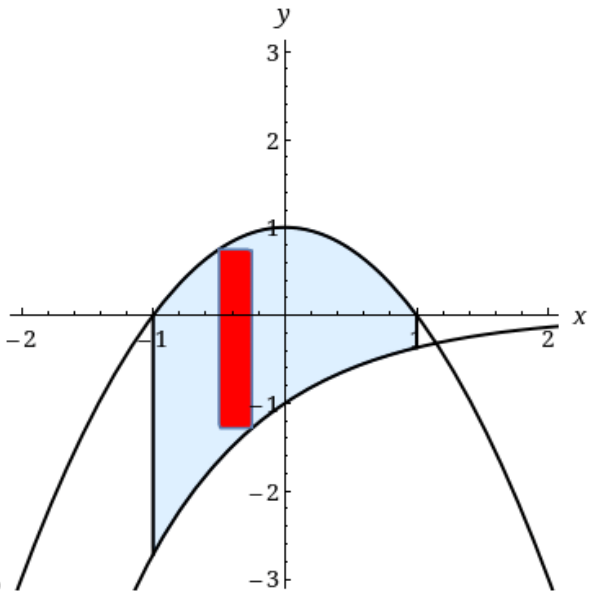
MY NOTES ASK YOUR TEACHER

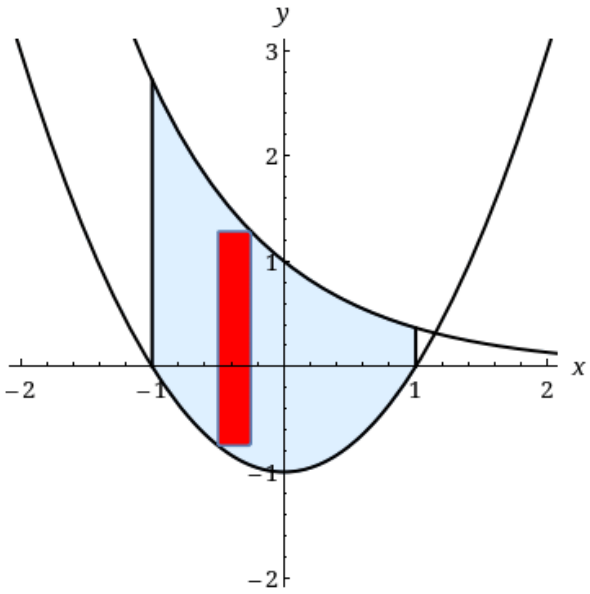
Sketch the region enclosed by the given curves. Decide whether to integrate with respect to x or y. Draw a typical approximating rectangle.

$y = e^x,\ y = x^2 - 1,\ x = -1,\ x = 1$









Find the area of the region.

$$(3e1−3e−1+4)3

Need Help? Read It

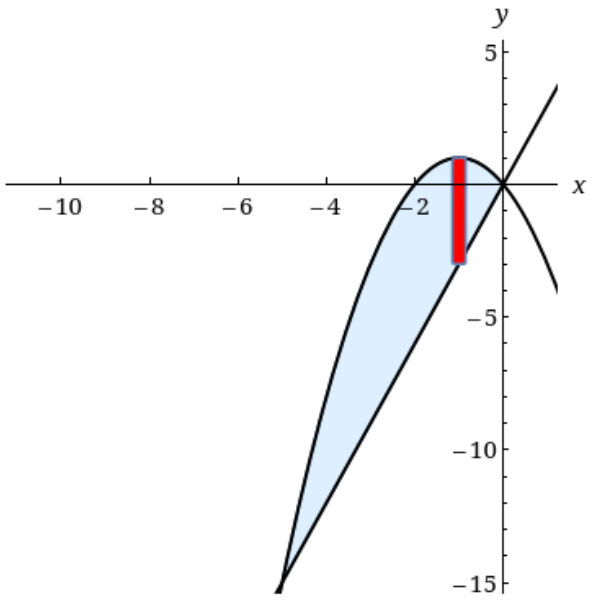
7. **[2/2 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 6.1.008.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

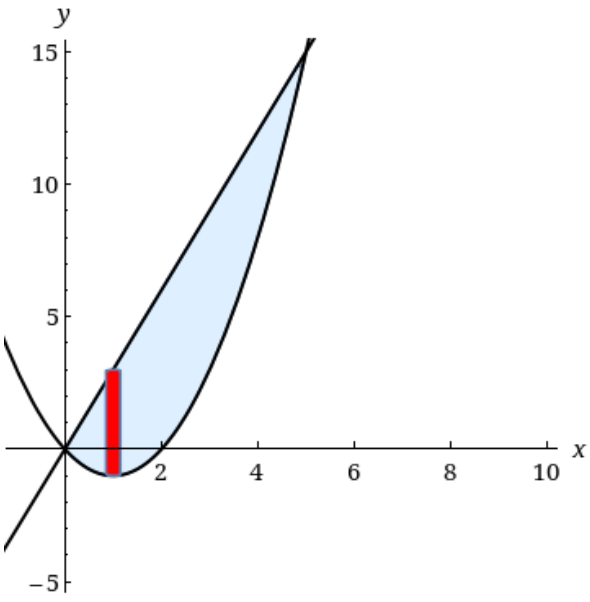
Sketch the region enclosed by the given curves. Decide whether to integrate with respect to x or y. Draw a typical approximating rectangle.

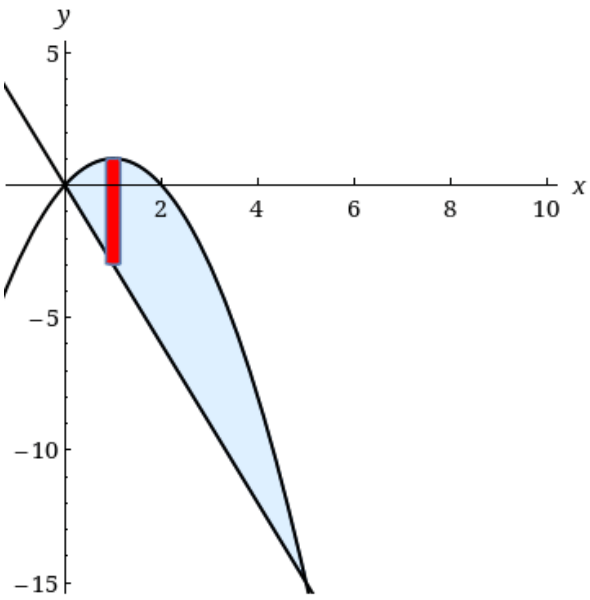
$y = x^2 - 2x,\ y = 3x$

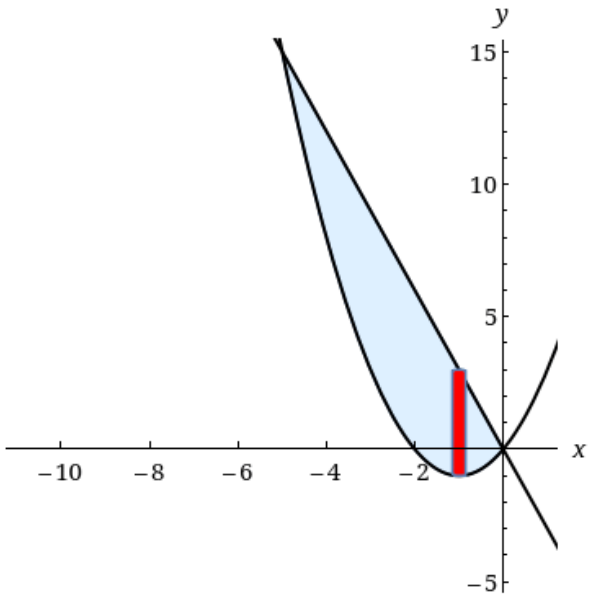
○



◉







Find the area of the region.

$$1256

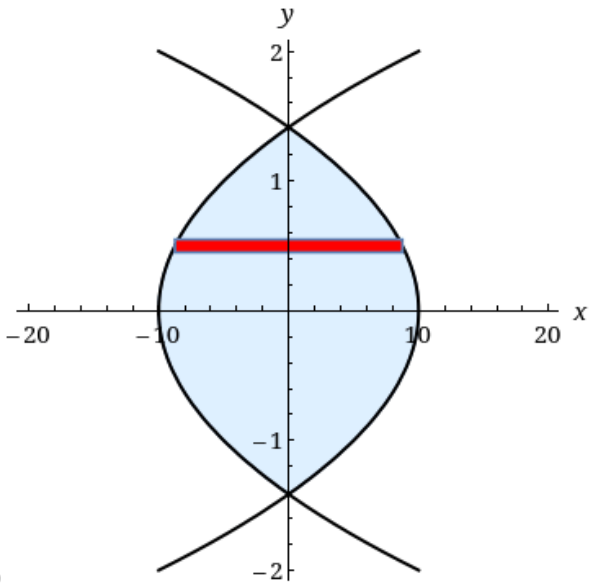
Need Help? Read It

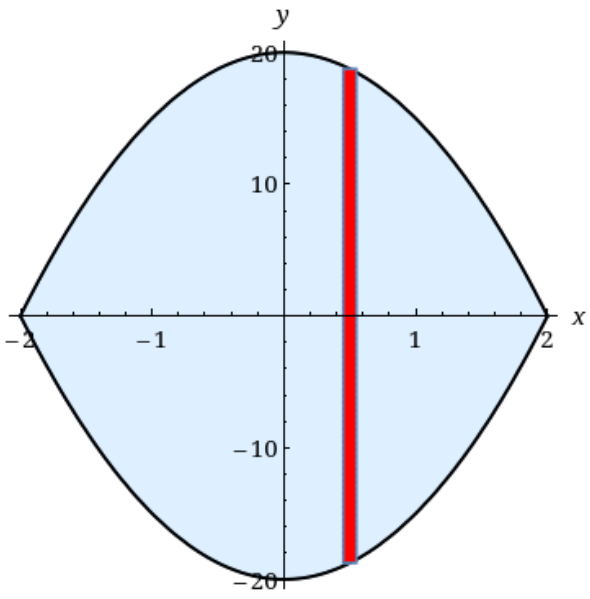
8. **[2/2 Points]** DETAILS PREVIOUS ANSWERS SCALCET8 6.1.011.MI.

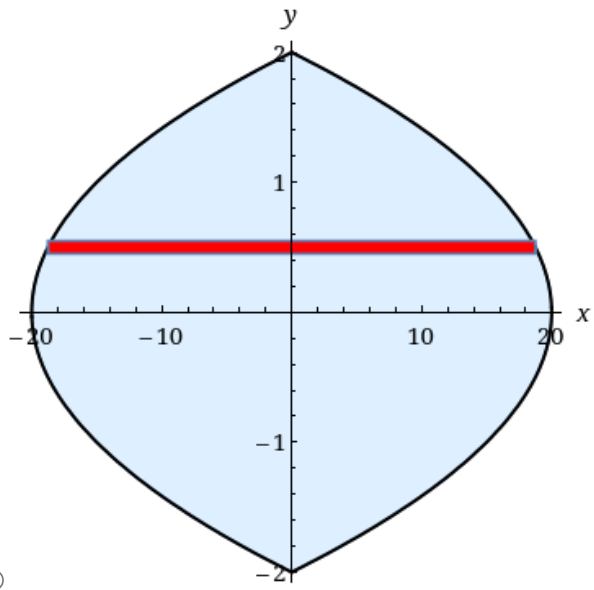
MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

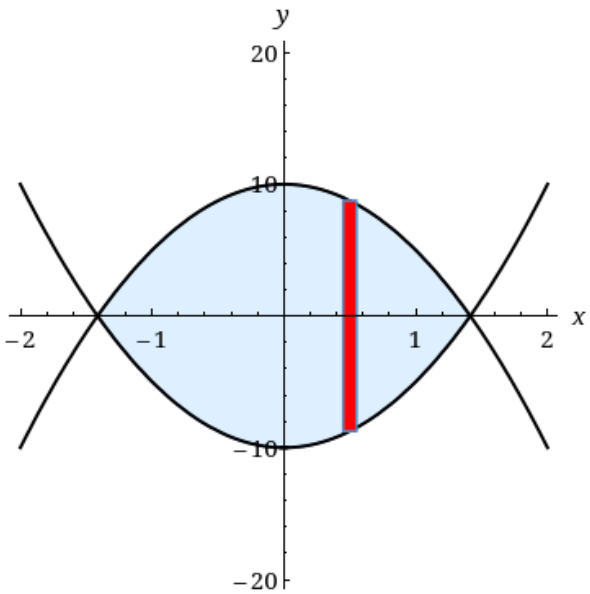
Sketch the region enclosed by the given curves. Decide whether to integrate with respect to x or y. Draw a typical approximating rectangle.

$x = 20 - 5y^2, \quad x = 5y^2 - 20$









Find the area of the region.

$$3203

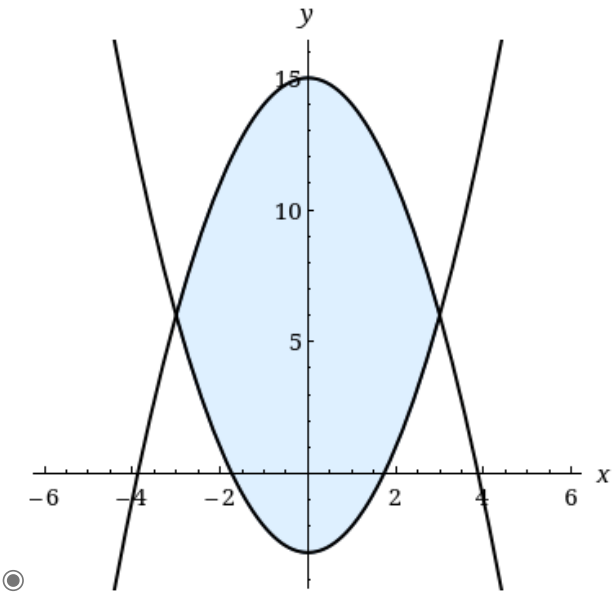
Need Help? Read It Watch It Master It

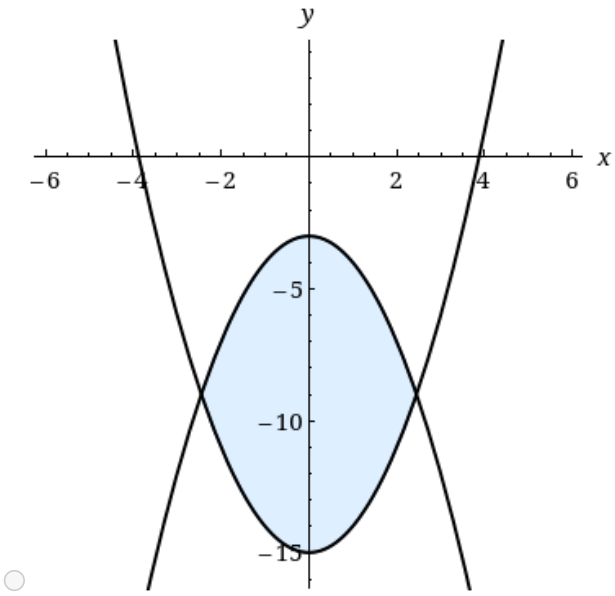
9. **[2/2 Points]** DETAILS PREVIOUS ANSWERS SCALCET8 6.1.013.

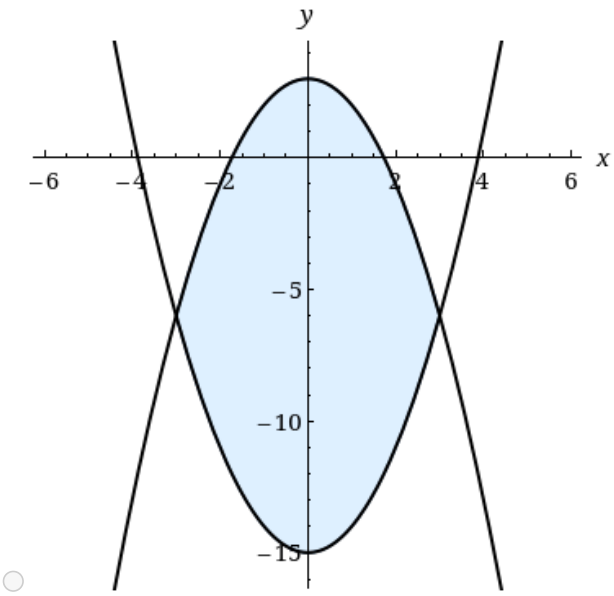
MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

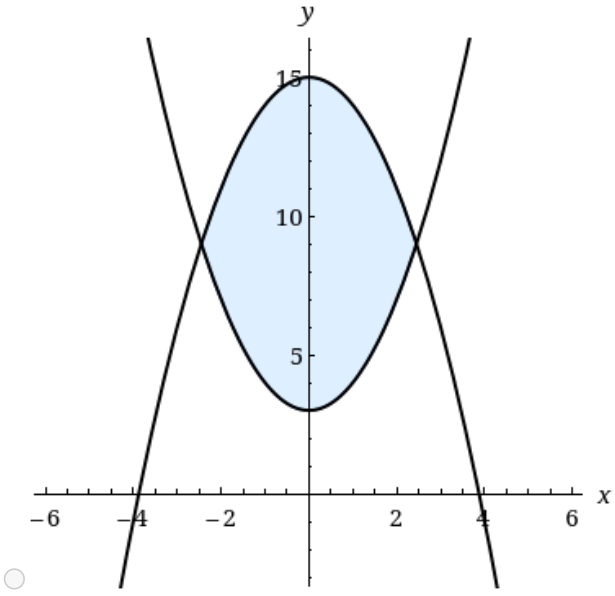
Sketch the region enclosed by the given curves.

$y = 15 - x^2, \quad y = x^2 - 3$









Find its area.

$$72

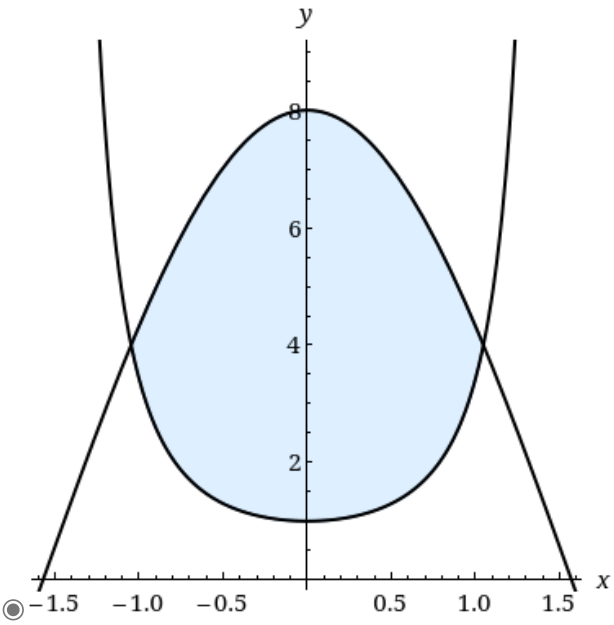
Need Help? Read It Watch It

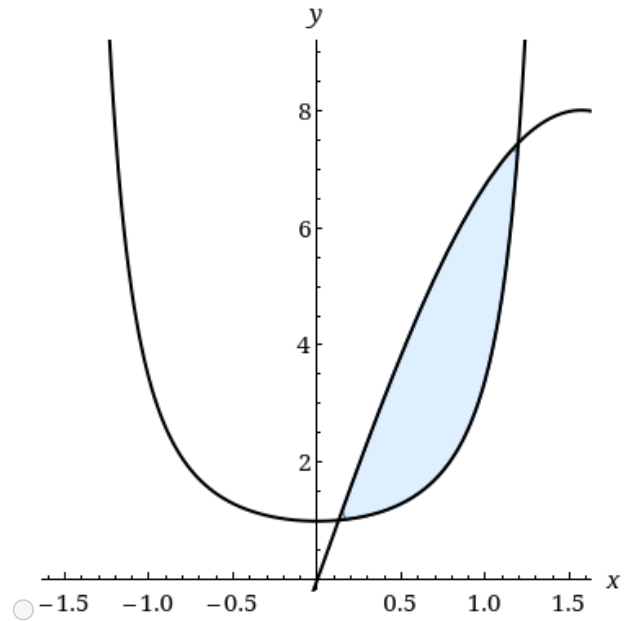
10. **[2/2 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 6.1.015.**

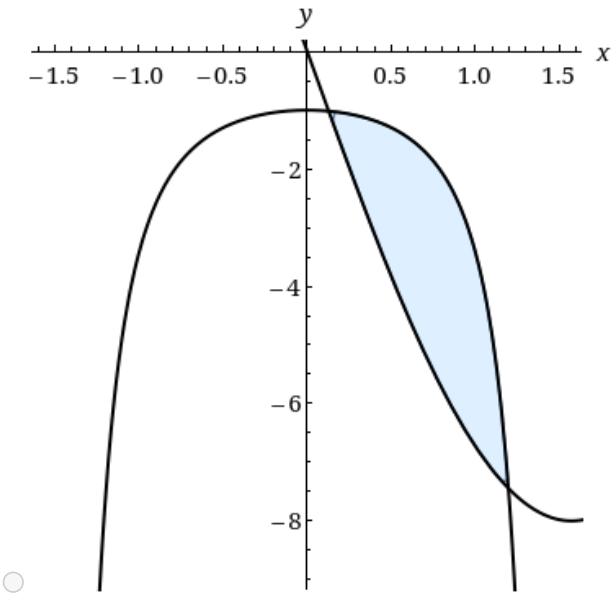
MY NOTES ASK YOUR TEACHER

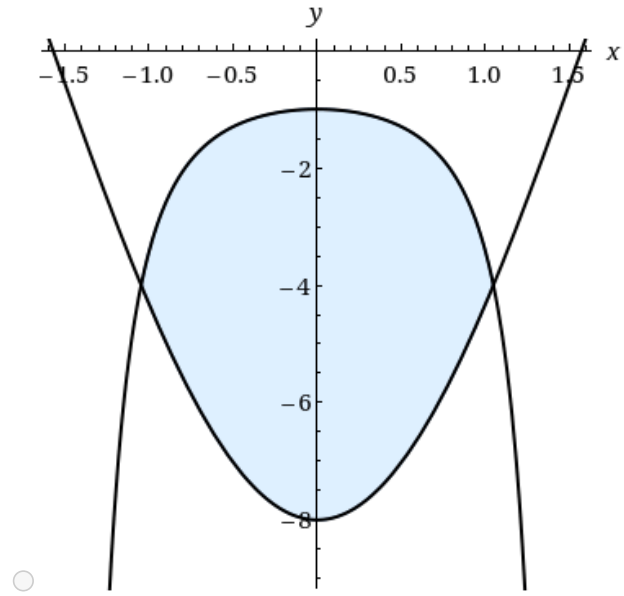
Sketch the region enclosed by the given curves.

$y = \sec^2(x), \quad y = 8\cos(x), \quad -\pi/3 \le x \le \pi/3$









Find its area.

$$6\sqrt{3}$$

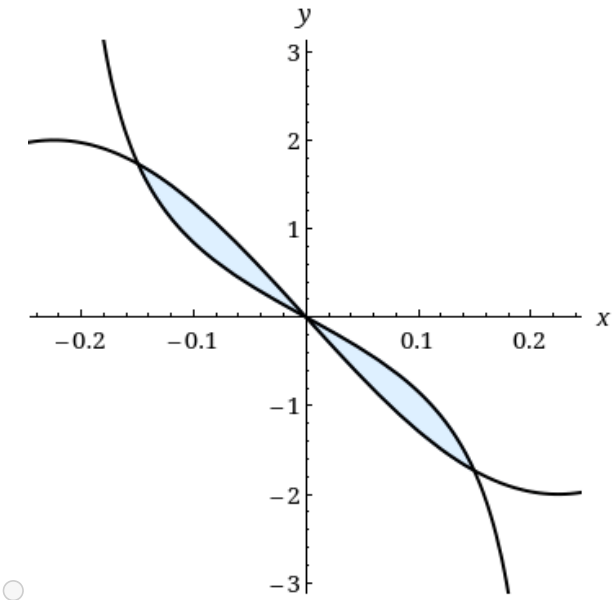
Need Help? Read It

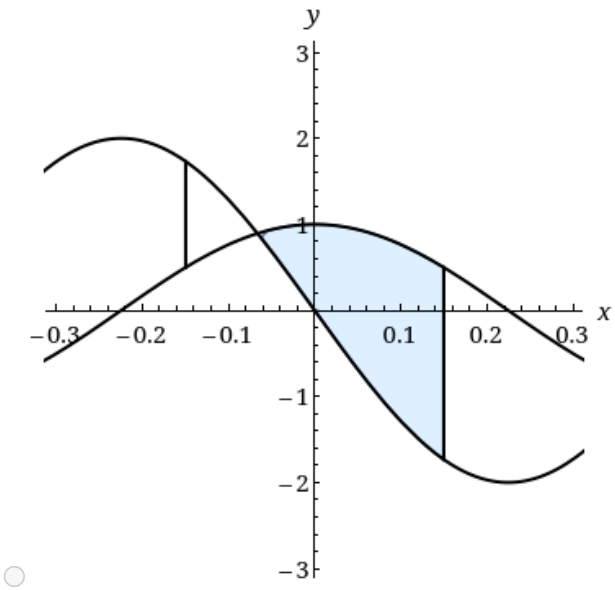
11. **[2/2 Points]** DETAILS PREVIOUS ANSWERS SCALCET8 6.1.021.

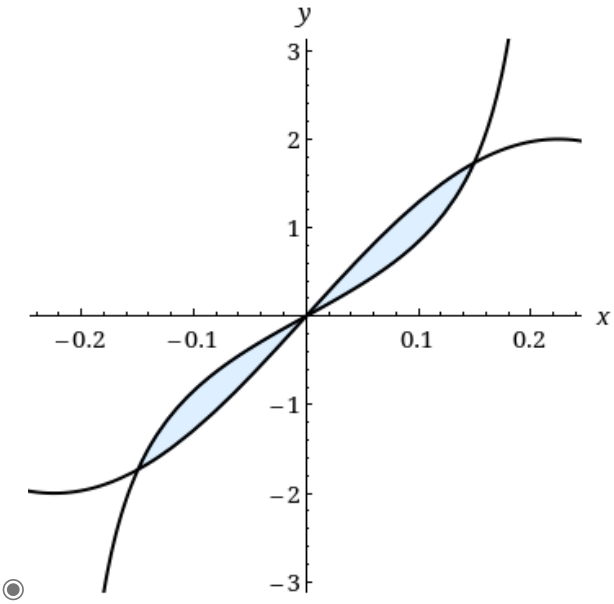
MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

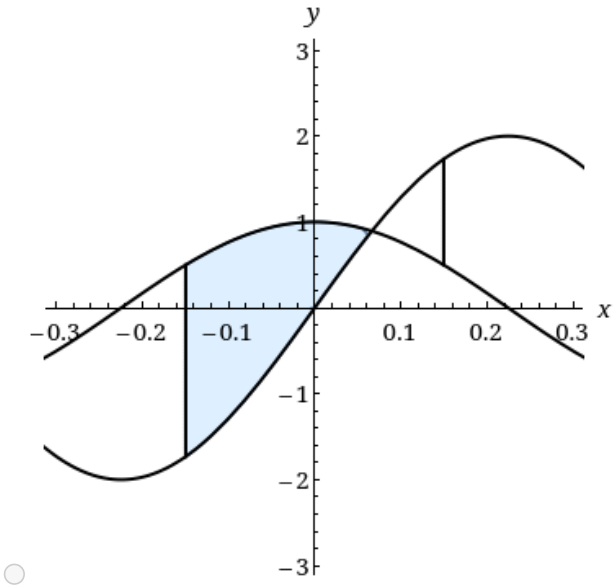
Sketch the region enclosed by the given curves.

$y = \tan(7x), \quad y = 2\sin(7x), \quad -\pi/21 \le x \le \pi/21$









Find its area.

$$(2−2ln(2))7

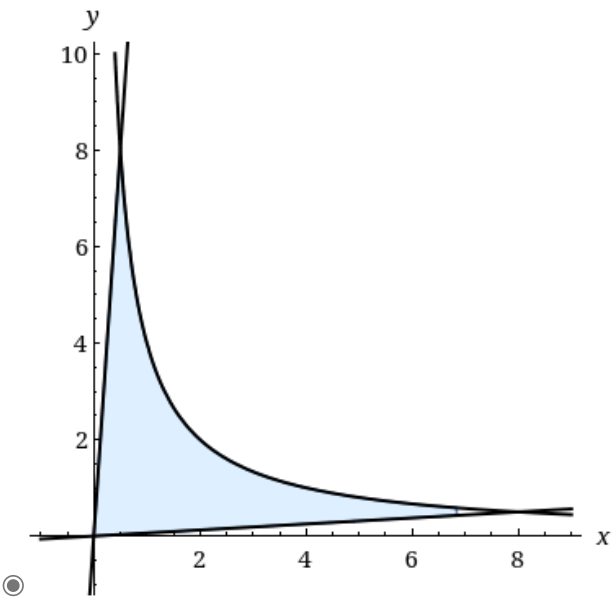
Need Help? Read It Watch It

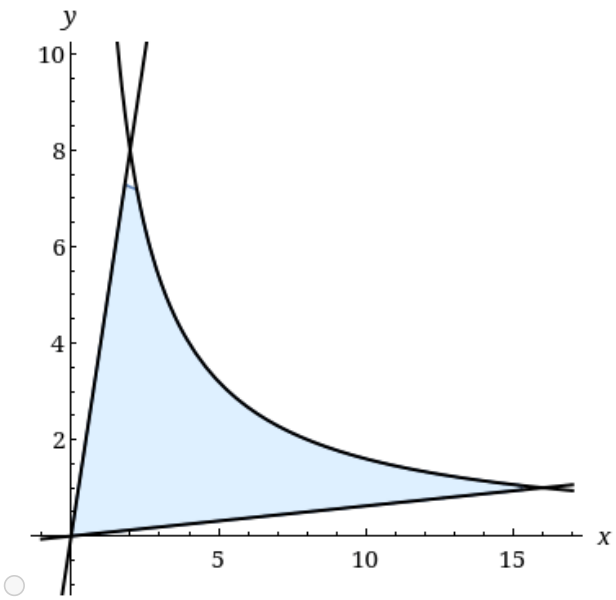
12. **[2/2 Points]** DETAILS PREVIOUS ANSWERS SCALCET8 6.1.027.

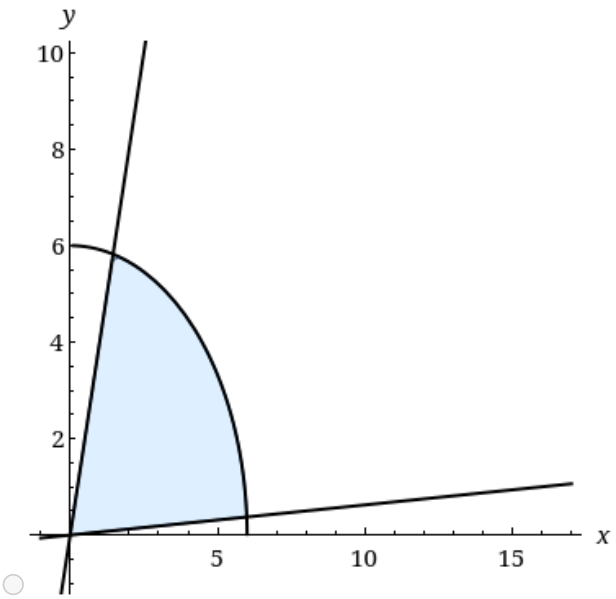
MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

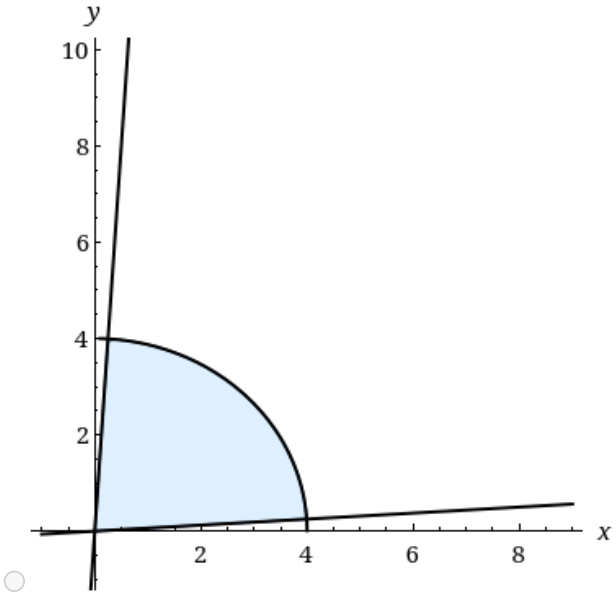
Sketch the region enclosed by the given curves.

$y = 4/x, \quad y = 16x, \quad y = \frac{1}{16}x, \quad x > 0$









Find its area.

$$4*ln(16)$$

Need Help? Read It Watch It

13. **[2/2 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 6.1.035.**

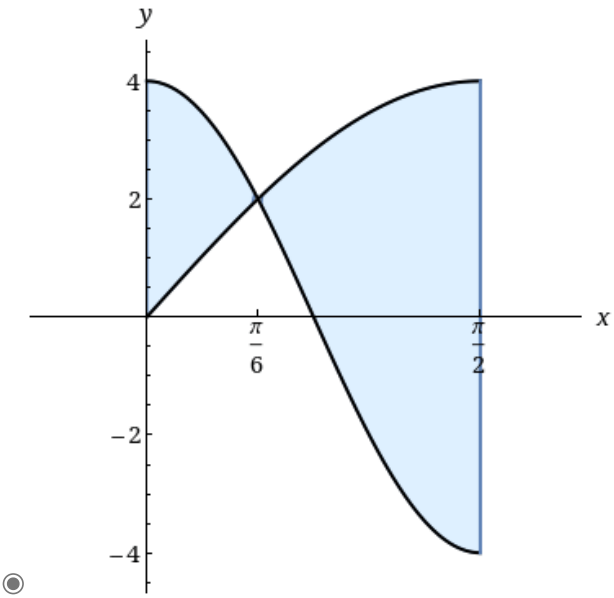
MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

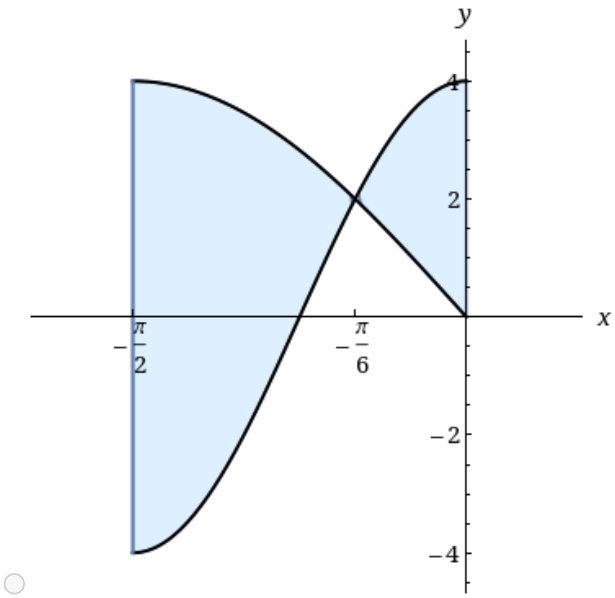
Evaluate the integral and interpret it as the area of a region.

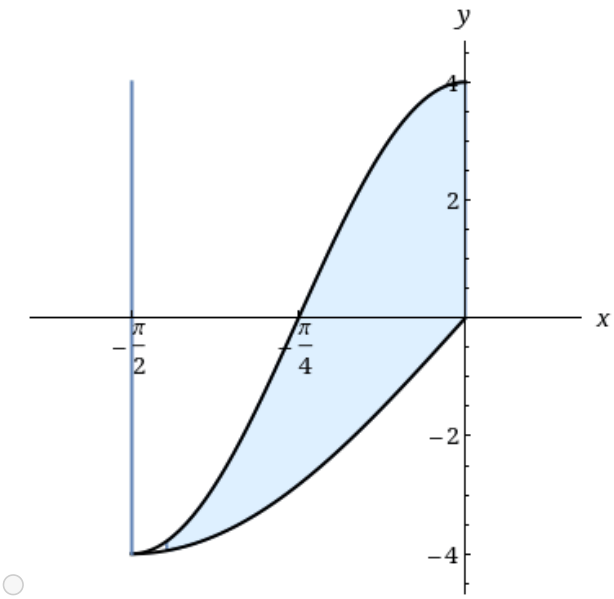
$$\int_0^{\pi/2} |4\sin(x) - 4\cos(2x)|\,dx$$

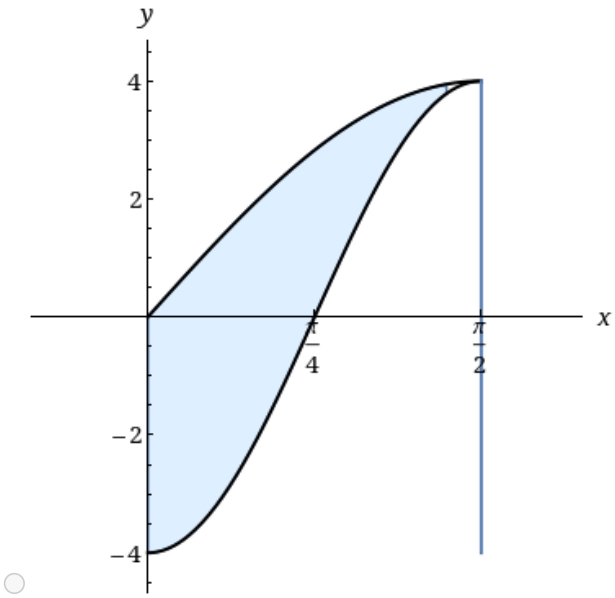
$$6\sqrt{3}-4$$

Sketch the region.









Need Help? Read It Watch It

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14. [1/1 Points] DETAILS PREVIOUS ANSWERS SCALCET8 6.1.056.

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Find the area of the region bounded by the parabola $y = 2x^2$, the tangent line to this parabola at (1, 2), and the x-axis.

1/6 ✔

Need Help? Read It

15. **[6/6 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 6.1.EI.001.**

MY NOTES ASK YOUR TEACHER

Review the Explore It, then use it to complete the exercise below.

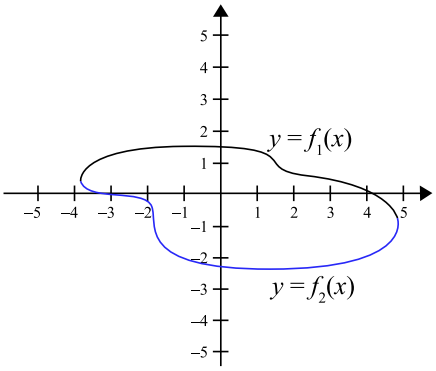
Explore It

CONCEPT EXAMPLE VIDEO EXPLORE & TEST WHEN WOULD I USE THIS

YOU WILL LEARN ABOUT:

How to determine the area between two curves by integrating with respect to x or y, which is a generalization of the concept of finding the area between a curve and the x-axis $(y = 0)$.





The area between the continuous curves

- $y = f_1(x)$ and $y = f_2(x)$ and between $x = a$ and $x = b$ is given by $\int_a^b [f_1(x) - f_2(x)]\,dx$, where $f_1(x) = f_2(x)$ for all x between a and b.
- $x = g_1(y)$ and $x = g_2(y)$ and between $y = c$ and $y = d$ is given by $\int_d^c [g_1(y) - g_2(y)]\,dy$, where $g_1(y) = g_2(y)$ for all y between c and d.

Click here to access the Explore It in a new window.

Select Function 1 under the Explore & Test section of the Explore It.

When integrating with respect to the *x*-axis, the formula for the approximate area, using *n* rectangles of equal width and left end points is

$$\text{Area} \approx \sum_{i=1}^{n} (\sqrt{x_{i-1}} - x_{i-1}^2)\,\Delta x.$$

(a) Set the number of rectangles in the simulation to 5. What is the approximate area? (Round your answer to five decimal places.)

0.30974 ✔

(b) Set the number of rectangles in the simulation to 10. What is the approximate area? (Round your answer to five decimal places.)

0.32551 ✔

(c) What is the formula, in terms of *n*, for the width Δ*x* of each rectangle? Before submitting your answer, verify your answer with various numbers of rectangles in the simulation.

$\Delta x =$
$$1n

(d) What is the formula, in terms of i and n, for the general left end point, x_{i-1}? Before submitting your answer, verify your answer with various numbers of rectangles in the simulation.

$x_{i-1} =$
$$(i-1)n

(e) Using your answers to parts (c) and (d), give the general formula in terms of i and n for the approximate area when integrating with respect to the x-axis using n rectangles of equal width and left end points.

$$Area \approx \sum_{i=1}^{n} ($$
$$(\sqrt{(i-1)n-((i-1)n)2)(1n)}$$
$$)$$

(f) Set the number of rectangles to 2. As you increase the number of rectangles to 5, 10, 20, and 40, which rectangles give the better approximation, vertical rectangles (integration with respect to the x-axis) or horizontal rectangles (integration with respect to the y-axis)?

- vertical rectangles
- horizontal rectangles
- sometimes vertical and sometimes horizontal
- (●) They give the same approximation.

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